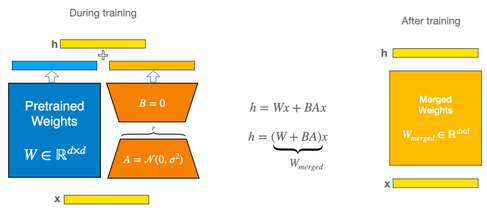
Re-Implementation of “LoRA: Low Rank Adaption of Large Language Models”

Ajay Tadinada (at663), Dan Wei Zuo (dz267), and Mericel Tao (mst223)
Cornell University, CS 4782 Final Project

Introduction

Full Fine-Tuning (FFT) of Large Language Models (LLMs) demands significant VRAM and extended training times.



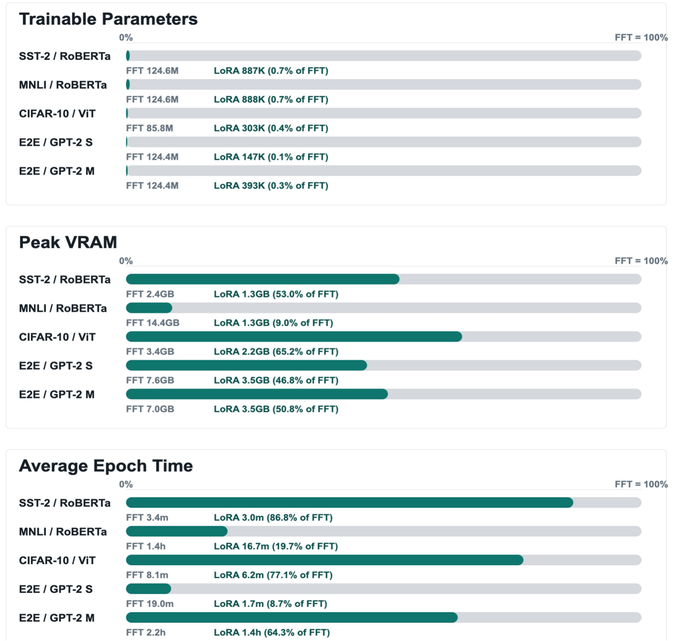
We aim to reduce trainable parameters and peak VRAM usage on a variety of transformer models while maintaining FFT accuracy.

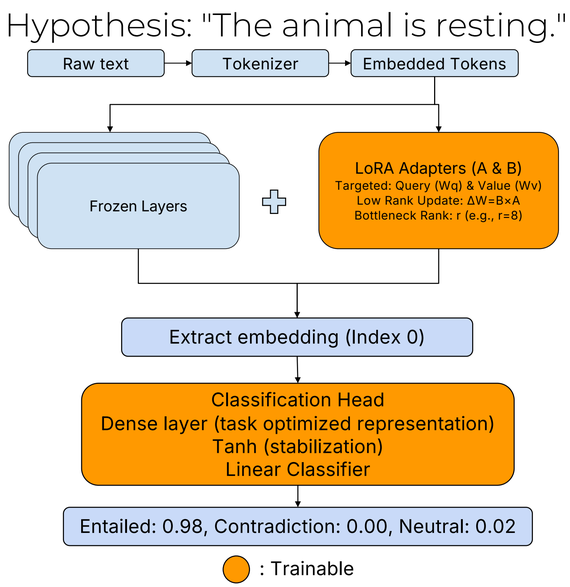
Methodology

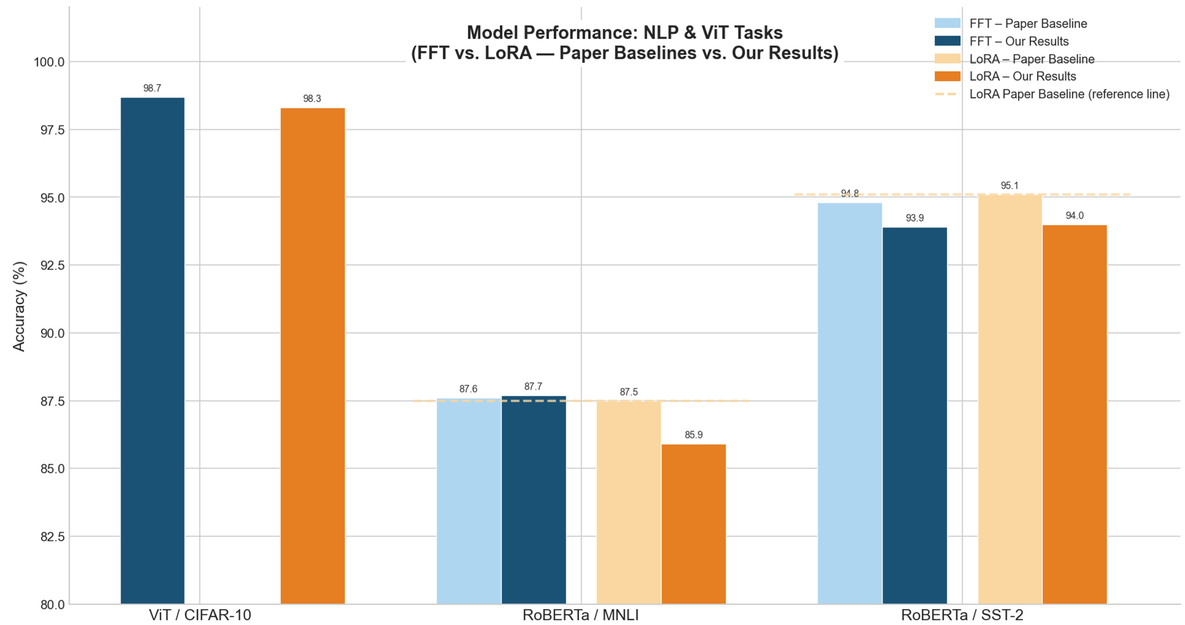
	RoBERTa	GPT-2	ViT
Hardware	A100	T4	T4
Experiment Type	Full FT vs LoRA	Small vs Medium	Generalization Test
Benchmark Dataset	SST-2 / MNLI	E2E NLG	CIFAR-10

Modifications:

- Focus on smaller models
- Adjusting hyperparameters for speed
- Extended experiments to explore LoRA’s generalizability to images

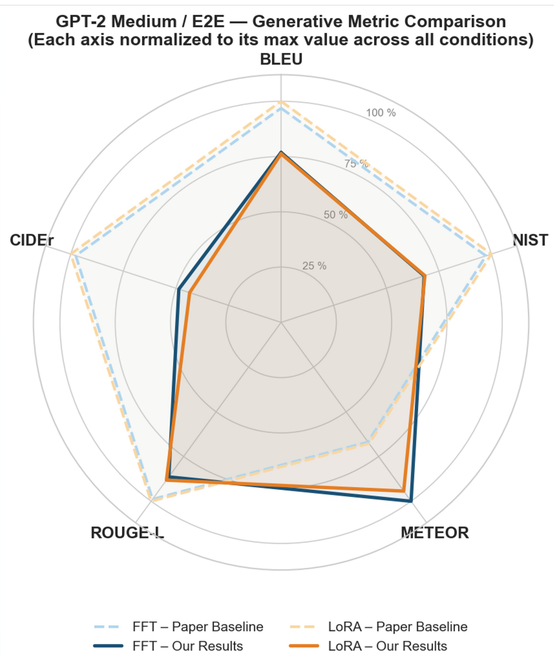






Conclusion

- Reduced trainable parameters by ~99%.
- Preserved accuracy on classification tasks
- Reduced peak VRAM and training time
- Possible sensitivity to task complexity, model scale, and hyperparameter tuning.
- LoRA is a practical, parameter efficient fine-tuning method when compute or memory is limited



Note: The METEOR metric is likely reporting higher scores than reality because our implementation reports the score compared to only the first reference.

References

(1) Edward J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models.” In: CoRR abs/2106.09685 (2021). arXiv:2106.09685. URL: https://arxiv.org/abs/2106.09685.

(1) Yinhan Liu et al. “RoBERTa: A Robustly Optimized BERT Pretraining Approach.” arXiv:1907.11692, 2019. URL: vhttps://arxiv.org/abs/1907.11692.

(3) Alec Radford et al. “Language Models are Unsupervised Multitask Learners.” OpenAI, 2019. URL: https://cdn.openai.com/better-language-models /language_models_are_unsupervised_multitask_learners.pdf.

(4) Alexey Dosovitskiy et al. “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.” In: International Conference on Learning Representations (CLR), 2021. arXiv:2010.11929. URL: https://arxiv.org/abs/2010.11929.

(5) Alex Wang et al. “GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding.” In: International Conference on Learning Representations (ICLR), 2019. arXiv:1804.07461. URL: https://arxiv.org/abs/1804.07461.

(6) Richard Socher et al. “Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank.” In: Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2013, pp. 1631–1642. URL: https://aclanthology.org/D13-1170/.